PyGameMaker — Tutorial 11: 2.5D First Steps

How to Use This Handout

This handout goes with the in-app tutorial **Help > Tutorials > 2.5D: First Steps** (4 pages, about 20-25 minutes). This lesson is not in the beginner edition: your teacher may need to switch the edition first. Keep the tutorial open on one half of your screen. Tick each box when the step is done, and check the **You should see** line at the end of every phase. If something does not match, look at **Stuck?** before you call your teacher.

What You Will Make

A small room you explore **in the first person**, like the classic game Wolfenstein 3D. It is still an ordinary 2D room (walls have a normal x and y); only the picture on the screen is drawn as if you were standing inside. You turn with the arrow keys, walk forward and back, and bump into real walls.

Phase 1: The Room and the Walls

- ☐ **1.** Create a project FirstPerson with the **Empty Project** template.
- ☐ **2.** Create sprite spr_wall (**32×32**, grey or bricks) and sprite spr_player (16×16, any colour; you will never see it).
- ☐ **3.** Create obj_wall (sprite spr_wall) and tick **Solid**. Create obj_player (sprite spr_player), **not** solid.
- ☐ **4.** Create room_main with width and height **320**, and show the grid (32 by default, so blocks snap).
- ☐ **5.** Place walls in every border cell: top row x 0 to 288 (y 0), bottom row (y 288), left and right columns between the corners, and three blocks inside at (128, 128), (160, 128) and (128, 160).
- ☐ **6.** Place obj_player at (48, 48).

> **You should see:** a 10×10 grid of cells with a ring of wall blocks and three blocks inside. (You cannot see the 3D view yet.)

Phase 2: The Camera and the Controls

- ☐ **7.** In obj_player, **When created**: *Enable Raycast View* with Field of View **66**, Cell Size **32**, Render Distance **20**. Leave Camera Object empty (it means "this object").
- ☐ **8. Keyboard: Left Arrow (held)**: *Set Facing Angle* **3**, Relative on. **Right Arrow (held)**: *Set Facing Angle* **-3**, Relative on.
- ☐ **9. Up Arrow (held)**: *Set Direction & Speed* direction facing_angle, speed **3**. **Down Arrow (held)**: direction facing_angle+180, speed **3**. **No key**: direction 0, speed 0.
- ☐ **10.** Add **Collision with obj_wall** and leave it **empty**. Do not skip it: without it the player walks through the walls.

> **You should see:** press **F5**. The room appears in first person: grey walls, a blue ceiling and a dark floor. The arrows turn and walk you, and you cannot leave the room.

Phase 3: Test and Tune

- [ ] **11.** Change one setting of *Enable Raycast View* at a time and press **F5**: Field of View 40 and 100; Render Distance 3; Wall Color, Floor Color, Ceiling Color; Columns.
- [ ] **12.** Write down what each change did in your notes.

> **You should see:** a narrow view for 40, a wide, slightly fish-eyed view for 100, far walls disappearing for a short Render Distance, and new colours.

Stuck?

What is wrong	Most likely cause	What to do
I see the normal top-down room	The *Enable Raycast View* action is missing, or it is on an object that is not in the room	Put it in obj_player's **When created** and place the player in the room
I walk through walls	The empty **Collision with obj_wall** is missing, or obj_wall is not Solid	Add the event (leave it empty); tick **Solid**
The walls look wrong or have gaps	A wall is off the 32-pixel grid, or Cell Size is not 32	Use grid snapping; set Cell Size 32
I turn the wrong way	The 3 and -3 are swapped	Swap them on the two turning events
I cannot move but I can turn	Up and Down have no *Set Direction & Speed*, or the direction is not facing_angle	Check the direction text exactly
I keep walking after I let go	The **No key** event is missing	Add direction 0, speed 0
The window is tiny	The window takes the size of the room (320×320)	Make a bigger room, keeping walls on the grid
Nothing is visible on the walls	The wall sprite is fully transparent	Draw a filled 32×32 sprite

Challenges

- **Try this (5 minutes):** make a dungeon or a Mars base with the three colour settings.
- **Push further:** build a larger room with corridors (keep every wall on the 32-pixel grid).
- **Invent:** make the turning faster or slower by changing the 3 and -3.

Vocabulary

Term	What it means
2.5D (raycast)	A first-person picture drawn from a 2D room, as in Wolfenstein 3D
Camera	The object the view is drawn from (here, the player)
Facing angle	The direction you look, in degrees (0 = right, 90 = up)
Field of view	How wide the view is, in degrees
Cell size	The size of one grid cell (32 pixels); walls must be on this grid

Check Yourself

Write your answers in the notes below.

1. Why is this called 2.5D and not 3D?
2. Why does the player need a **Collision with obj_wall** event that has no actions?
3. What does *Relative* mean when the facing angle is set?

My Notes